

♦ ADVANCED CONCRETE REINFORCEMENT

# Chopped Basalt Fibre

A high-performance reinforcing fibre that combines strength, thermal stability and chemical resistance — a cost-effective alternative to steel, glass and carbon fibre. With more than 70 million monofilaments per kilogram, basalt fibre binds concrete in three dimensions for dramatically improved strength and abrasion resistance.

**2.5×**

TENSILE STRENGTH VS STEEL

**~1/3**

THE WEIGHT OF STEEL

**700°**

MAX OPERATING TEMP (°C)

**0%**

CORROSION

— WHY BASALT FIBRE

## Benefits in Concrete

Mixed into concrete, asphalt, resin or plastic, basalt fibre reinforces from within — lifting strength and durability while cutting long-term maintenance.

— KEY BENEFITS

- 🔗 **Impact & Splitting Resistance**  
A three-dimensional fibre network resists dynamic loads, splitting and impact.
- 🔗 **Stronger Bond**  
Improves binding strength and load transfer through the concrete matrix.
- 🔗 **Abrasion & Wear Resistance**  
Stands up to heavy traffic — industrial floors, highways and hard-wearing surfaces.
- 🔗 **Fire & Thermal Stability**  
Non-combustible volcanic rock with a melting point near 1,450 °C.
- 🔗 **Chemical, Alkali & Frost Resistance**  
Performs in aggressive, marine and freeze-thaw environments.
- 🔗 **Less Cracking, No Delamination**  
Restrains shrinkage cracks during curing and keeps the mix homogeneous.



♦ LEARN MORE

### Scan for the Full Product Range

Detailed specifications, case studies and technical datasheets.

[www.basaltechsa.com](http://www.basaltechsa.com)



— DIRECT ECONOMIC EFFECT

- ✓ Reduced thickness at equal strength
- ✓ Replaces steel rebar — less labour
- ✓ Longer service life, lower upkeep
- ✓ Lighter — lower transport cost

# Properties & Specification

Basalt fibre is stronger than steel and polypropylene fibre, and stretches less than glass fibre — so loaded concrete cracks and shrinks less, and stays intact for longer.

## FIBRE COMPARISON

| PROPERTY                      | BASALT ♦ | PP      | GLASS       | STEEL     |
|-------------------------------|----------|---------|-------------|-----------|
| Tensile (MPa)                 | 3500     | 150–600 | 1500–3500   | 600–1500  |
| Diameter                      | 13–17µm  | 10–25µm | 13–15µm     | 0.5–1.2mm |
| Length (mm)                   | 24       | 18      | 18          | 40        |
| Elasticity (GPa)              | ≥ 75     | 35      | 75          | 190       |
| Elongation (%)                | 3.2      | 20–150  | 4.5         | 3–4       |
| Melting (°C)                  | 1450     | 160     | 860         | 1550      |
| Alkali / Corrosion Resistance | High     | High    | Alkali only | Low       |
| Density (g/cm³)               | 2.6      | 0.91    | 2.6         | 7.8       |

Indicative raw-fibre values. Steel column = carbon-steel fibre.

## ENGINEERED COATING

A water-based PVC coating (carboxylate styrene-butadiene latex) improves chemical bonding with cement, adhesion to aggregates, mechanical stability and water resistance.



## RECOMMENDED DOSAGE

| CONCRETE TYPE / APPLICATION               | CHOPPED FIBRE        |
|-------------------------------------------|----------------------|
| Industrial floors, treated base           | 1% cement wt.*       |
| Industrial floors, heavy load (untreated) | 1% cement wt. + mesh |
| Columns, RC frames, spacers               | 1.5 kg/m³            |
| Building & structure foundations          | 1.5–2 kg/m³          |
| Road construction                         | 1% cement wt.*       |
| Render / plaster                          | 0.6–1% cement wt.    |
| Foam / aerated concrete                   | 0.25–0.5% cement wt. |

\*Replaces metal rebar. ~1 kg/m³ controls micro-cracking; higher-class concrete needs more, up to a saturation point.

## MIXING TIP

Mix the fibres dry with sand, stone and cement, then add water. Extend mixing by 30–45 seconds for even dispersion. Beyond the saturation point, extra fibre stops adding strength.

## APPLICATIONS

- ✓ Industrial floors & highways
- ✓ Earthquake-resistant structures
- ✓ Airstrips & military bases
- ✓ Corrosive-liquid concrete pipes
- ✓ Asphalt — less rutting, better fatigue life

## OUR LOCATIONS

**G GAUTENG**  
Plot 112, Oak Tree R563  
Mogale City, Gauteng

**L LIMPOPO**  
118 OR Tambo Drive  
Lephalale, Limpopo

## DIRECT CONTACTS

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## WHY BASALTECHSA

**Proudly South African**  
Local knowledge, world-class BFRP standards. Serving Gauteng & Limpopo.

**Expert Technical Support**  
From specification to mix design, our team provides end-to-end guidance.



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